Project Idea: Intelligent Intrusion Prevention System (IPS)

Our project focuses on building an Intelligent Intrusion Prevention System (IPS) that combines network monitoring, data analysis, and artificial intelligence to detect and prevent cyberattacks in real time.

The main concept is to design a simulated environment where: - Victim servers are intentionally exposed to different types of attacks (such as DoS, port scanning, brute force, etc.). - Monitoring tools like Security Onion are deployed to capture all incoming and outgoing traffic (PCAP files). - The collected raw traffic data is processed and transformed into structured formats (CSV/JSON).

Once the data is prepared, it will be used to train a Machine Learning model capable of distinguishing between normal user activity and malicious traffic patterns. This AI-powered model will then be integrated into a backend API that can automatically analyze network flows and return alerts indicating whether a potential intrusion is happening.

To make the system more practical and user-friendly, the alerts and detection results will be displayed on a Dashboard interface, allowing administrators to visualize attacks in real time and take quick action.

The entire solution will be containerized using Docker and deployed through a CI/CD pipeline on a cloud provider such as AWS or Azure. This ensures scalability, automation, and easy maintenance of the IPS.

Additionally, a Penetration Testing team will attempt to attack the deployed system in order to validate its effectiveness, while an Incident Response team will handle logs, alerts, and produce detailed reports about detected intrusions.

In summary, the project brings together networking, security, AI, backend development, frontend dashboards, DevOps practices, and cybersecurity testing into one integrated IPS platform.

IPS Project - Detailed Team Roles Guide

1. Networking Team
- Build the virtual environment (Ubuntu victim, Kali attacker, Security Onion monitor).
- Connect all VMs inside the same internal network.
- Capture traffic (normal + attack) and export as PCAP files.
- Deliver PCAP files to AI Team.

2. AI Team
- Convert PCAP to structured data (CSV/JSON).
- Train a Machine Learning model (e.g., Random Forest).
- Export trained model as .pkl file.
- Deliver model + preprocessing scripts to Backend Team.

3. Backend Team (.NET Core)
- Develop ASP.NET Core Web API.
- Integrate AI model into API.
- Create endpoint /detect for attack detection.
- Deliver working API to Frontend + DevOps.

4. Frontend Team
- Build Dashboard (React/Vue.js).
- Display detected attacks and alerts in real-time.
- Integrate with Backend API.
- Deliver frontend code to DevOps.

5. DevOps Team
- Containerize all components (Backend, Frontend, etc.).
- Deploy on AWS/Azure using Docker and GitHub Actions.
- Ensure CI/CD pipeline is working.
- Deliver final running system link to Pentest team.

6. Penetration Testing Team
- Attack the deployed system using various tools (Nmap, SQLmap, Flood attacks).
- Write report about weaknesses and suggestions.
- Deliver report to Incident Response Team.

7. Incident Response Team
- Monitor alerts and verify real vs false alarms.
- Collect evidence of incidents.
- Write detailed incident reports.
- Deliver reports to management.

Flow: Networking → AI → Backend → Frontend → DevOps → Pentesting → Incident Response

Team Integration Diagram


Networking
AI
Backend
Frontend
DevOps